
Measuring Cities with New Technologies

LSE GY457 - AT 2025

Topic 10

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Course overview

- Block 1

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- Topic 2: The empirics of agglomeration
- Topic 3: The costs and benefits of agglomeration

- Block 2

- Topic 4&5: Monocentric city model
- Topic 6: Firm location choice

- Block 3

- Topic 7: The vertical dimension of cities
- Topic 8: Neighborhoods and segregation
- Topic 9: Gentrification and suburbanization
- **Topic 10: Measuring cities with new technologies**

Recap from last time

- Determinant of dynamic spatial income pattern
 - New expensive transportation mode pushes rich to suburbs
 - Lifecycle of building supply
 - Consumption amenities attract young college-graduate to centers
 - Gentrifiers attracted to rich neighborhoods due to spillover effects
- Welfare effects of gentrification
 - Renters lose unless benefit from spillovers or adjust preferences
 - Homeowners benefit except for loss of social capital
- Evidence points to limited displacement effects and benefits to stayers
 - Stayers may still be net losers if they rent

Today: Measuring cities with new technologies

- Introduction
 - Improving understanding through measurement
- Detecting urban expansion and land use patterns
- Mapping accessibility and human mobility
- Spatial externalities: Congestion, health, and the environment
- Challenges and the future of cities
 - Implications for the future of real estate

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Old vs. new data sources

- Old: Surveys and records
 - Field surveys, census records, ...
- New: Human activity
 - Social network, mobility data, cell phone records,...
- New: Built environment
 - Remote sensing (aerial, satellite imagery), street-level imagery

Pros and cons?

Remote sensing in Economics

Table 1
Main Satellite Data Sources Used in Economics

<i>Source</i>	<i>Economics applications</i>	<i>Highest resolution</i>	<i>Pricing</i>	<i>Availability by year</i>	<i>Examples</i>	<i>For more information</i>
Landsat	Urban land cover, beaches, forest cover, mineral deposits	30 m	Free	1972– (8 satellites)	Foster and Rosenzweig (2003); Faber and Gaubert (2015)	http://landsat.usgs.gov/
MODIS	Airborne pollution, fish abundance	250 m	Free	1999– (Terra); 2002– (Aqua)	Foster, Gutierrez, and Kumar (2009); Burgess et al. (2012)	http://modis.gsfc.nasa.gov/
Night lights (DMSP-OLS, VIIRS)	Income, electricity use	~1 km	Free digital annual (DMSP- OLS) and monthly (VIIRS) composites	Digital archive 1992–2013+ (VIIRS 2012–; film archive 1972–1991)	Henderson et al. (2012); Chen and Nordhaus (2011)	http://ngdc.noaa.gov/eog/
SRTM	Elevation, terrain roughness	30 m	Free	2000 (static)	Costinot, Donaldson, and Smith (2016) via Global Agro-Ecological Zones (GAEZ) data	https://lta.cr.usgs.gov/SRTM
DigitalGlobe (including Quickbird, Ikonos)	Urban land cover, forests	< 1 m	Not free	1999– (6 satellites)	Marx, Stoker, and Suri (2015); Jayachandran, de Laet, Lambin, and Stanton (2016)	https://www.digitalglobe.com/resources/satellite-information

Source:
Donaldson and
Storeygard (2016)

“Big data” solves some but not all issues (Glaeser et al., 2016)

- City data available and usable at high temporal frequencies and high levels of spatial granularity
- Big data gives finer picture of urban life
- But issues related to causal inference remain
 - Still need to find sources of exogenous variation
 - In urban economics, often tied as “shocks to people” or “shocks to places”
 - e.g., Moving To Opportunity experiment (*shock to people*)
 - e.g., Opening of new metro stop/improvements in commuting technology (*shock to places*)

Urban questions and big data

(Glaeser et al., 2016)

A *How does urban development influence the economy?*

- Administrative income data (e.g., through tax sources)
- Geographic and fine level of detail reduces measurement error (improves precision) but likely makes sorting more problematic
 - Sorting within cities more severe than sorting across cities
 - *“The most natural explanation of why some denser zip codes pay higher wages (despite offering workers the same housing costs and city-level amenities) is that those zip codes have more able workers.”*
 - Combine with shocks to places (e.g., new infrastructure policies)

Urban questions and big data

(Glaeser et al., 2016)

- B *How does the physical city interact with social outcomes?*
- Big data makes it possible to measure city characteristics at finer levels (street level, physical geo)
 - Improvements on social outcomes too (e.g., social network, mobility data, crime, health, education)

Urban questions and big data

(Glaeser et al., 2016)

C *How much do people value urban amenities?*

- Assess the benefits of urban amenities with hedonic models that use home sales by address, record quality of restaurants through online platforms, measures of urban consumption
 - But in itself big data does not help to run better identified regressions

Urban questions and big data

(Glaeser et al., 2016)

- *How can public policy improve the quality of physical space?*
- Big data may help managing cities more effectively
(local measurement of crime, geolocation of potholes,...)
- *Other ideas?*

“Shock to people”: The Moving to Opportunity experiment (Chetty et al., 2016)

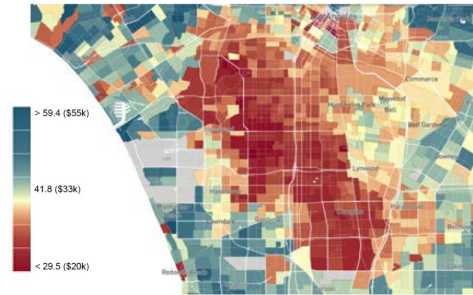
- MTO: Poorer households were randomly allocated vouchers that enabled and incentivized them to move to lower-poverty areas
- Children who were moved early enough experienced clear improvements in earnings
- IRS records (income tax returns) allowed to follow participants over time
- ...and to build the [*Opportunity Atlas*](#) (Chetty et al., 2018)
 - Dataset that provides estimates of long-term outcomes of children born in each neighborhood in the US

Opportunity Atlas (Chetty et al., 2018)

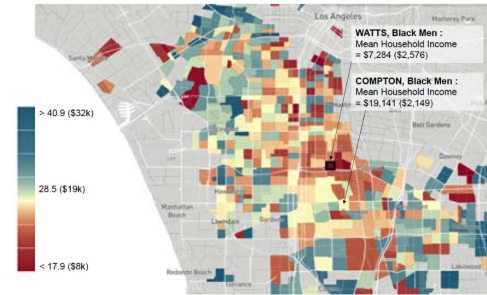
- Census tract level statistics using individual level panel dataset from US Census Bureau covering (nearly) the entire American population from 1989 to 2015
- Link parents to children's outcomes in adulthood
 - 20.5 million children in the 1978-1983 birth cohort
- In each tract-by-gender-by-race cell, estimate the conditional expectation of children's outcomes given parents household income
- Neighborhood matter at very granular level
 - Tract characteristics 1 mile away matter little for upward mobility

Opportunity Atlas (Chetty et al., 2018)

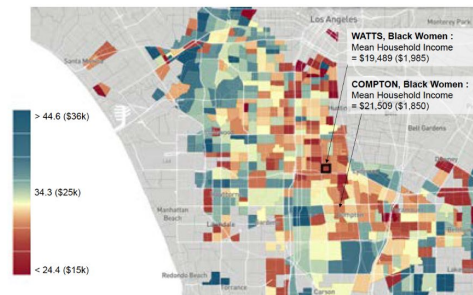
A. All Children: Household Income Given Parents at 25th Percentile



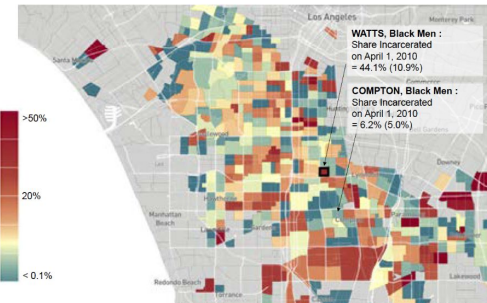
B. Black Men: Household Income Given Parents at 25th Percentile



C. Black Women: Household Income Given Parents at 25th Percentile



D. Black Men: Incarceration Rates Given Parents at 1st Percentile



“Big data” solves some but not all issues (Glaeser et al., 2016)

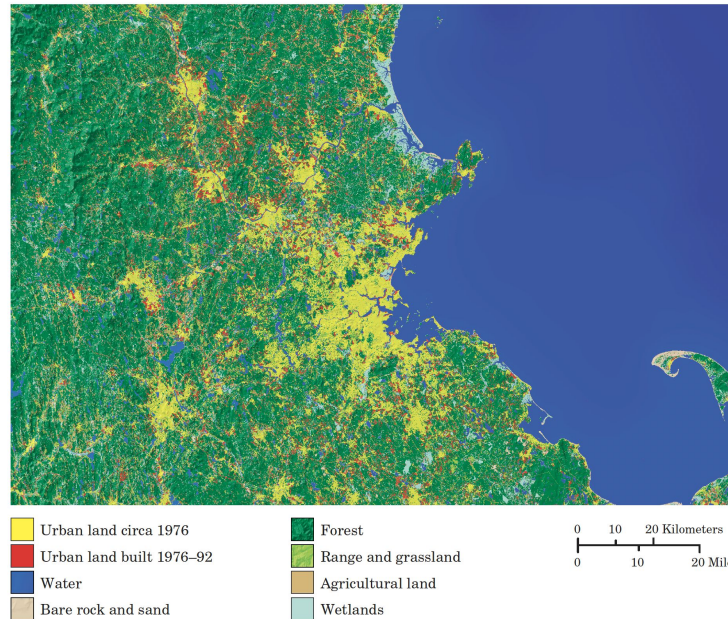
“In general, big data can meaningfully help big cities and improve research on cities—but only if it is used with thoughtful care. Big data will do far more for urban research if it is paired with exogenous sources of variation. Big data will do far more for policymaking and implementation if it is paired with openness to new methods. As in most areas, new technology works best if it is accompanied by human capital.”

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Causes of sprawl: A portrait from space

(Burchfield et al., 2006)



- Seminal study of the determinants of urban sprawl
- Use remote sensing data to track evolution of land use on a grid of 8.7 billion 30 x 30 meter cells in US
- Sprawl measured as amount of undeveloped land surrounding average urban dwelling

Causes of sprawl: A portrait from space

(Burchfield et al., 2006)

SPRAWL INDICES FOR METROPOLITAN AREAS WITH POPULATION OVER ONE MILLION

Metropolitan area	Sprawl index for 1992 residential land	Sprawl index for 1976 residential land	Metropolitan area	Sprawl index for 1992 residential land	Sprawl index for 1976 residential land
Atlanta	55.57	57.77	Minneapolis-St. Paul	32.07	31.34
Boston	47.64	44.72	New Haven	39.11	38.68
Buffalo	39.92	37.87	New Orleans	32.29	33.92
Charlotte	52.73	51.12	New York	28.75	28.47
Chicago	31.76	31.21	Norfolk	40.82	44.07
Cincinnati	47.79	47.45	Orlando	40.02	39.39
Cleveland	36.84	36.24	Philadelphia	42.51	43.03
Columbus	41.20	41.59	Phoenix	27.54	34.94
Dallas	28.08	26.65	Pittsburgh	57.70	56.71
Denver	28.63	28.63	Portland	44.90	43.38
Detroit	33.28	30.47	Rochester	48.80	48.11
Greensboro	52.94	51.45	Sacramento	34.93	30.72
Hartford	41.34	42.23	Salt Lake City	31.90	32.88
Houston	38.15	38.93	San Antonio	32.77	29.58
Indianapolis	39.66	37.68	San Diego	45.63	45.40
Kansas City	35.32	34.33	San Francisco	30.48	29.81
Los Angeles	35.41	32.95	Seattle	46.97	45.03
Memphis	27.40	28.72	St. Louis	43.44	40.62
Miami	20.73	20.03	Tampa	36.01	34.84
Milwaukee	35.33	33.85	Washington-Baltimore	49.81	50.68

Causes of sprawl: A portrait from space

(Burchfield et al., 2006)

- Urban models normally treat space as featureless plain
- But natural and other features affect urban sprawl:
 1. Car-dependency (vs. public transit)
 2. Slow population growth
 3. Uncertainty around population growth
 4. Specialization in sectors characterized by higher employment decentralization
 5. Rugged terrain, no high mountains nearby, aquifers in urban fringe, temperate climate

Cities in bad shape (Harari, 2020)



FIGURE 1. SHAPE METRICS: AN EXAMPLE

- Causal economic effects of city shape in India
- City footprint relevant for transit accessibility and service delivery
 - Use satellite imagery and historical maps
- Compact cities have faster pop. growth and negative compensating wage differentials (higher QoL)

Mumbai mills redevelopments and spatial spillovers (Gechter and Tsivanidis, 2023)

(a) 2000



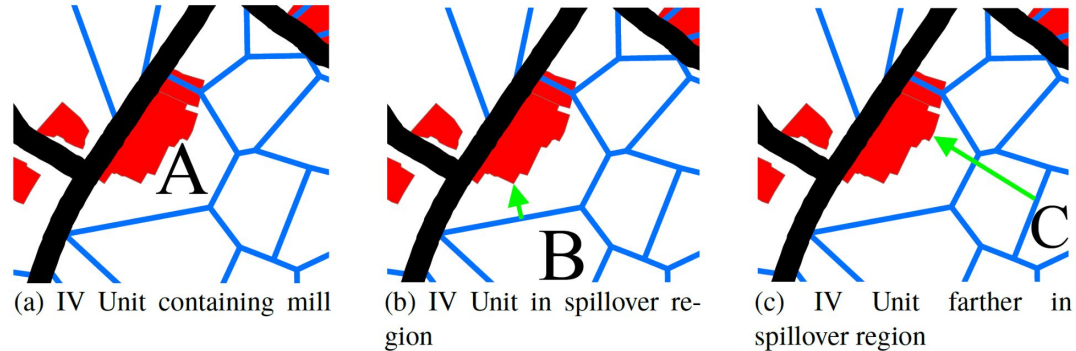
(b) 2016



- Study local spillovers from high-rise developments
- Most of Mumbai's 60 textile mills (covering 15% central city land) redeveloped into high-rises in the 2000s

Mumbai mills redevelopments and spatial spillovers (Gechter and Tsivanidis, 2023)

Figure 2: Measuring treatment from mill redevelopment



Notes: Mill sites are represented by red areas and IV Unit boundaries are in blue (black lines represent city ward boundaries). The IV Unit containing the letter “A” in panel 2a belongs in category (1) because there is a mill site inside it. The IV Unit containing the letter “B” in panel 2b belongs in category (2) because it does not have a mill site inside it and the distance between its boundary and the closest mill site, represented by the green arrow, is 500m or less. The IV Unit containing the letter “C” in panel 2c has a greater distance to the nearest mill boundary because it does not contain a mill and the green line is larger than the one in panel 2b

Mumbai mills redevelopments and spatial spillovers (Gechter and Tsivanidis, 2023)

- Threats to identification could lead to over- or understatement of the true effect
- Show no pre-trends in floorspace prices between treatment and control units
 - Suggest no anticipation and no selection
- Find sizable local spillovers consistent with new developments gentrifying the surrounding neighborhoods

Mumbai mills redevelopments and spatial spillovers (Gechter and Tsivanidis, 2023)

Table 1: Main Results

	Res Price	Slum Share	Inf Emp	Form Emp	Ed Index	Population
Contains Mill	0.130** (0.059)	-0.226** (0.102)	-0.413** (0.186)	0.196 (0.330)	0.262** (0.126)	-0.150** (0.059)
1m-500m	0.089* (0.049)	-0.270*** (0.097)	-0.400** (0.201)	0.202 (0.286)	-0.197 (0.175)	-0.117* (0.061)
500m-1000m	0.046 (0.048)	0.084 (0.109)	0.298 (0.258)	0.286 (0.365)	-0.157 (0.168)	-0.111 (0.092)
<i>N</i>	1,050	894	486	438	446	434

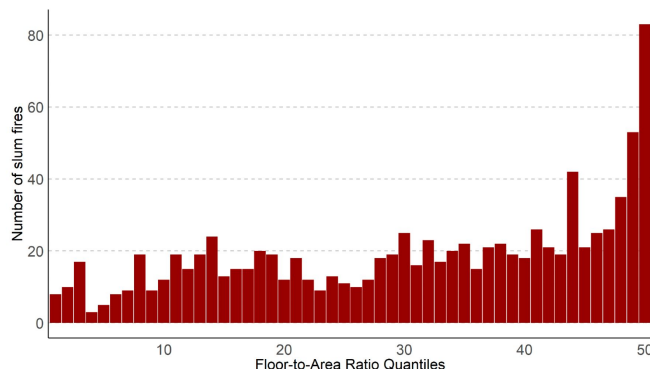
Notes: Robust standard errors in parentheses. Dependent variables are logged with the exception of slum share which is estimated by Poisson Pseudo Maximum Likelihood, and education index which is in standard deviations of the 2002 education index. Regressions include own spatial unit fixed effects and allow for differential time trends by log distance to CBD, log spatial unit area, log of 1+initial period road density, initial slum share, log of initial employment density, log of initial population density, and initial education index. To guard against endogeneity generated by serially correlated errors we do not allow differential trends by lagged outcome values. We trim 1% tails of outcome variables. Each specification has two time periods, in most

Mumbai mills redevelopments and spatial spillovers (Gechter and Tsivanidis, 2023)

- What are the aggregate effects? Are there distributional consequences (who are the winners and losers)?
 - Answer developing a rich spatial model
- Findings:
 1. City-wide benefits (higher floorspace supply, lower prices, space for formal firms)
 2. Displaced (slum) residents heavily hurt (would need to be compensated x3 actual compensation policy)

To burn a slum (Pucci, 2024)

Figure 4: Number of fires by quantiles of average real formal property value around slums, Sao Paulo (2001-2016)



Notes: Quantiles are calculated separately for each year. For example, the same slum can be in quantile 20 in one year and 25 in another.

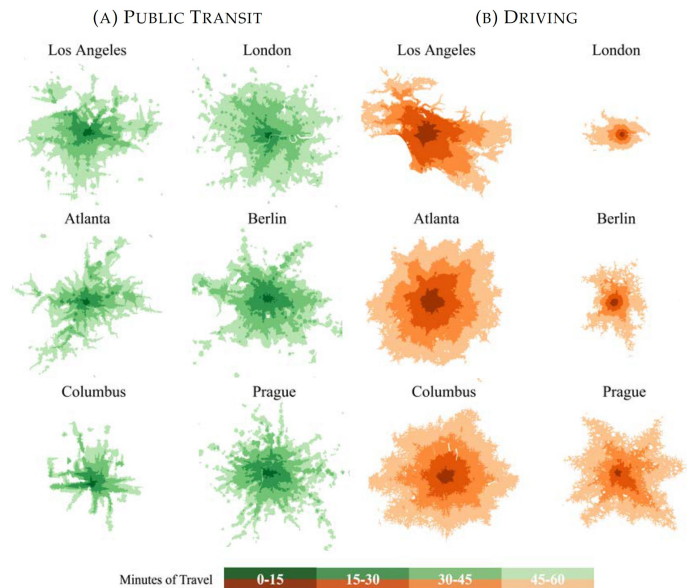
- Geocoded info on slums, fires, and land prices in Sao Paulo
- Examine the use of arson as violent tool to force slum removal in high-prices locations
- Show that probability of slum fires largely increases with land prices

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Measuring city center accessibility

(Conwell et al., 2023)



- Size of the surrounding area from which center can be reached within specific time
- Compute accessibility zones for US and European cities
- US cities are half as accessible via public transit than European cities, but twice as accessible by car

Measuring city center accessibility

- Similar tools may inform policies
 - e.g., Statistics Canada ([link](#)) on different types of urban amenities (accessibility to primary and secondary education, to health care facilities, places of employment, groceries, cultural venues, ...)
- ...and businesses alike (e.g., where to locate to maximize proximity to consumers, or where to build new developments)
- Paired with GPS data could allow to identify left-behind neighborhoods and target transit policies more effectively

The returns to face-to-face interactions

(Atkin et al., 2022)

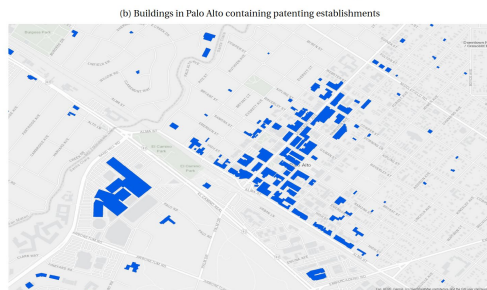
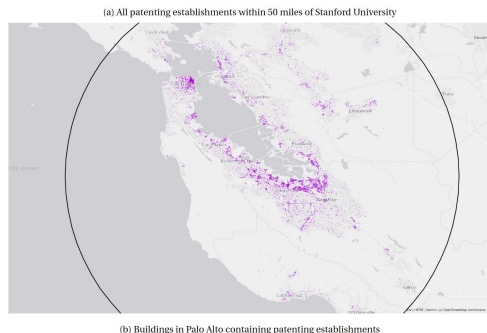
- Try to disentangle the effect of face-to-face interactions from other spatial spillovers
- Very hard to measure!!
 - Krugman (1991): *“knowledge flows are invisible, they leave no paper trail by which they may be measured and tracked”*
- Using granular smartphone geolocation data to measure face-to-face interactions between workers at different patenting firms in Silicon Valley
 - To study degree to which knowledge flows they measure citations among the firms

The returns to face-to-face interactions

(Atkin et al., 2022)

- Main data: Smartphone location pings from 50 million handsets ($\frac{1}{5}$ of US smartphone users) between 2016/17
- Smartphones report physical location every 5/10 min at accuracy of 20 meters
- For subset of data that belongs to workers, they observe pings for an average of 14.3h conditional on observing the phone in a day
 - Locate 51,000+ workers in a month
 - “Meeting” when two phones located close to each other at same time

The returns to face-to-face interactions (Atkin et al., 2022)



Notes: Figure 1a plots patenting establishments from Orbis located within 50 miles of Stanford University. Figure 1b marks rooftops of buildings in Palo Alto containing patenting establishments, using Microsoft's rooftop shapefile.

Figure 3: Overlay of pings for workers at Apple and Google

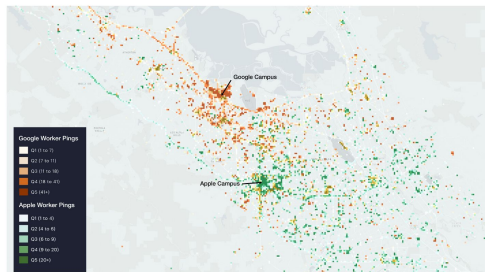
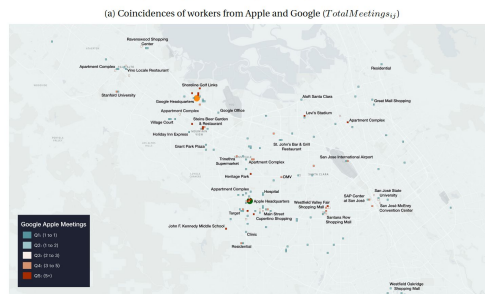


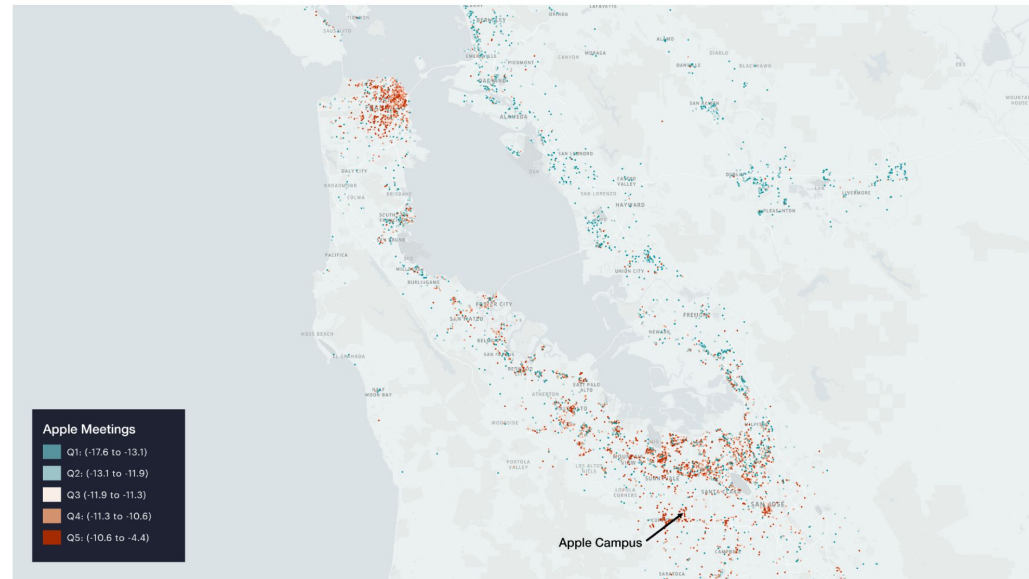
Figure 4: Apple and Google meetings measures



The returns to face-to-face interactions

(Atkin et al., 2022)

(a) $\ln TotalMP_{ij}$ variation between Apple and all other establishments



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The fast, the slow, and the congested

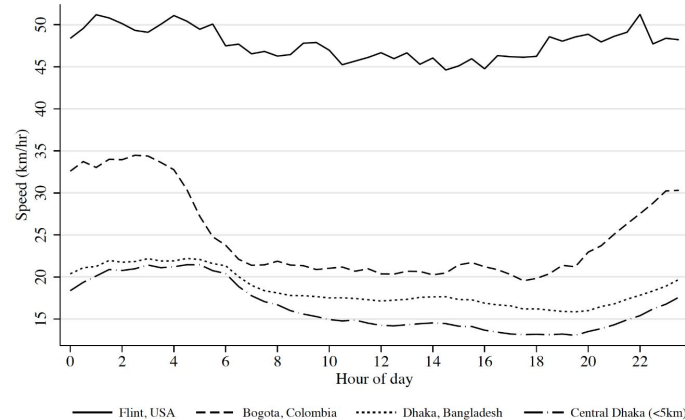
(Akbar et al., 2023)

- Data from a navigation app to examine the determinants of urban travel speed
 - Focus on explaining differences between rich and poor countries
- 1,200+ cities in 152 countries that contain 96% of the world's urban population excluding China
- More than 500 million trips simulated on Google Maps
- Most of the difference in city travel speed is *across countries*, not within countries

The fast, the slow, and the congested

(Akbar et al., 2023)

Figure 1: Travel speed throughout the day: The fast, the slow, and the congested

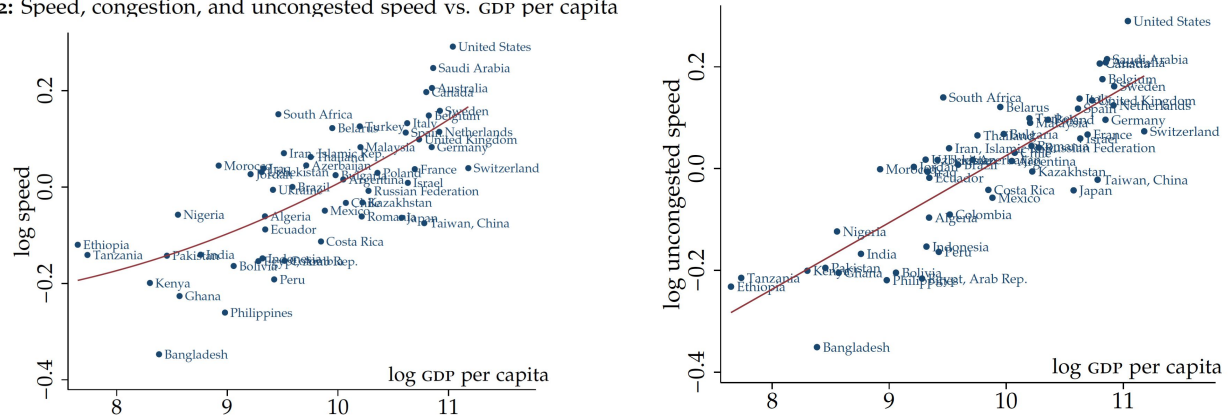


Mean speed for trips with length between 5 and 10 kilometers. Dhaka is the slowest, Flint is the fastest, and Bogota is the most congested city in the world. Central Dhaka refers to trips that take place on average within 5 kilometers of the center of Dhaka. See section 2 for further details.

The fast, the slow, and the congested

(Akbar et al., 2023)

Figure 2: Speed, congestion, and uncongested speed vs. GDP per capita



- Commuters in rich countries travel at a speed 50% higher on average than commuters in poor countries

The fast, the slow, and the congested

(Akbar et al., 2023)

- Large share of the difference can be explained by differences in national income per capita
- Build model to understand why urban travel faster in richer countries
 - Decomposition of size, infrastructure, and topography of the city
- Find that the “speed advantage” of rich vs poor country cities attributable to the type of infrastructure, i.e. more major (wider) roads
 - Operate increasing uncongested speed (not reducing congestion overall)

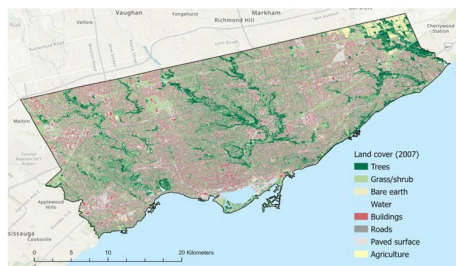
The fast, the slow, and the congested

(Akbar et al., 2023)

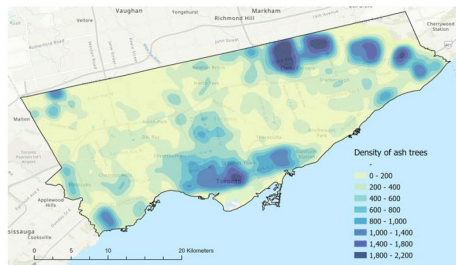
- US is faster because its cities have lower populations, wider areas, more major roads, and more grid-like road networks
- Bangladesh is slower because its cities are crossed more by water bodies, are more populous, and have fewer major roads

Cool cities (Han et al., 2024)

Figure 3. Land use classification in 2007 and (city-managed) ash trees.



(a) Land use (2007)



(b) City-managed ash trees

Notes: Panel (a) displays land use as produced by the Urban Tree Canopy (UTC) Assessment in 2007. Land use is divided into 8 categories: tree canopy (dark green), grass/shrub (lighter green), bare earth (sand), water (blue), buildings (red), roads (dark gray), other paved surfaces (light gray) and agriculture (yellow). Panel (b) shows the local density of city-managed ash trees, across 8 bins of density. Except for the central area, the neighborhoods of Scarborough (East), Mount Pleasant, North York and Etobicoke are the ones with the highest concentration of (city-managed) ash trees.

- Estimate the value of urban trees
 - Lower temperatures and save energy consumption
- Empirical strategy: assess the hedonic value of urban trees using residential property transactions
 - *Do we expect leafy neighborhoods to positively or negatively affect prices? Any threats to identification?*

Cool cities (Han et al., 2024)

- Thought experiment: Compare two identical neighborhoods with different levels of tree coverage
- Quasi-natural experiment: Leverage on an invasive beetle accidentally imported from abroad that started decimating ash trees in Toronto over time
- IV strategy: Instrument the evolution of tree coverage in a neighborhood by its exposure to the beetle infestation

Cool cities (Han et al., 2024)

Figure 4. Ash trees and the evolution of the tree canopy—an illustration.



Notes: This Figure shows the land use classification in a given neighborhood in the North-East of Toronto (James Park Square, Scarborough)—with a relatively high density of ash trees. The data was produced in 2007 (left panel) and 2018 (right panel) by Urban Forestry as part of an Urban Tree Canopy (UTC) Assessment. Land cover is represented by the following classes: tree canopy (dark green), grass/shrub (lighter green), bare earth (sand), water (dark blue), buildings (red), roads (dark gray), other paved surfaces (light gray) and agriculture (yellow). The green symbols represent the location of city-managed ash trees *at baseline*, as geolocated from their street addresses (Street Tree General Data, 2010). The latter explains why city-managed trees appear to be located within private lots. In our baseline specification, we aggregate tree cover at the postal code level, which mitigates the repercussions of such approximation.

- Data: comprehensive urban forest assessments of Toronto (2007/2018)
- ...and! unique geo-referenced register of all city-managed urban trees (to isolate exposure to the infestation across neighborhoods)
- Finding: 1 single tree adds 0.45% to property prices within a postal code

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Challenges and the future of cities

(Glaeser and Cutler, 2022)

- Centripetal transportation technologies of the 19th century promoted urban growth
- Centrifugal 20th century marked by the spread of innovations that enabled humans to disperse and cities to sprawl
- Enabling interactions seems to ensure the future of urban office space, but provides no guarantee for any individual city
 - Remote work and preferences/dislike to commute, and higher degrees of flexibility mean that cities will have to work harder

Challenges and the future of cities

(Glaeser and Cutler, 2022)

- Growing disconnect between home and workplace
 - Prediction that 20/25% of jobs will be remote
- If that holds true, new businesses could start almost anywhere
 - Attracted by low taxes, and low price destinations
- In cities with higher shares of remote jobs, demand for urban real estate will go down
 - Rents adjust and cities/spaces become more affordable
 - May the demand for city space fall so much to remain vacant?
And if so, what's next?

Summary

- New (granular) data sources make possible fundamental advancements in urban research
 - But need to be paired with exogenous sources of variation
- Ability to process satellite imagery and mobility data result in creative and original works to improve our understanding of the functioning of cities
- Fast-evolving urban dynamics open up new questions and challenges for businesses, policymakers, academics

Thank you!

Readings and references

Required readings

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